

CHAPTER - 2**HUMAN REPRODUCTION****SEXUAL DIMORPHISM IN HUMAN BEINGS**

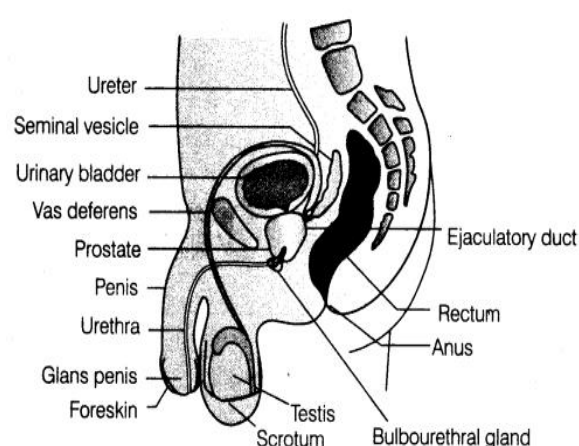
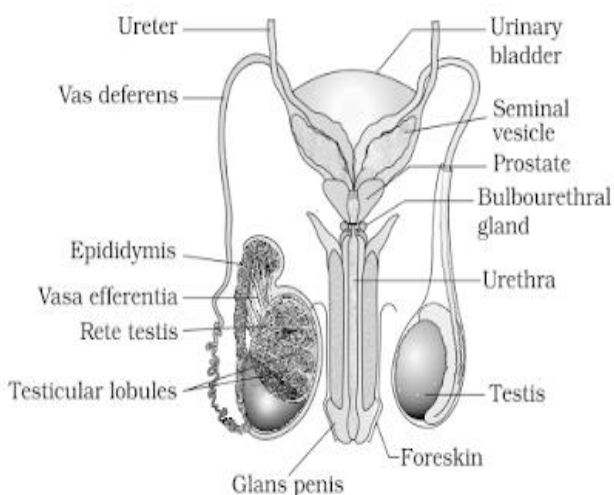
- Human beings, like most of mammals, are **unisexual** or **dioecious** (sexes are separate) sexually reproducing and viviparous.
- Each human being has only either male or female gonads, reproductive ducts and accessory genital structures.
- Out of these reproductive structures, gonads (testes in male and ovaries in females) are called **primary sex organs** and produce the gametes and sex hormones.
- All other genital structures (reproductive ducts and associated glands) are collectively called **secondary sex organs**. These include epididymis, vas deferens, prostate gland, seminal vesicles and penis in male, and the Fallopian tubes (oviducts), uterus, vagina and mammary glands in females.
- Two sexes are also different in their certain external morphological features, called **secondary** or **accessory sexual characters**, while this phenomenon is called **sexual dimorphism**.
- Sexual dimorphism can be studied by two types of sexual characters.
- After the child birth, primary sexual characters are mainly used to detect sex of a child. In males, primary sexual characters are a pair of testicles and a penis. In female, primary sexual characters are vaginal apparatus pair of ovaries and uterus.
- Once an individual reaches adolescence starts exhibiting secondary sexual characters.

Table showing Accessory sex characters in Man (Male) and Woman (Female)

CHARACTERS	MAN/MALE	WOMAN/FEMALE
Body build	More muscular and stronger	Less muscular and weaker
Hair	Beard, moustaches and chest hair present	Beard, moustaches and chest hair lacking
Skin	More hairy and coarse	Less hairy and soft
Voice	Hard and low pitch	Smooth and high pitch
Adam's apple (Larynx)	More prominent	Less prominent
Shoulder	Broad	Narrow
Pelvis/Hip region	Narrow	Broad
Sex hormones	Testosterone	Estrogen
Breasts (Mammary glands)	Undeveloped	Very well developed
Sex organs	Pair of testicles and penis	Pair of ovaries, uterus and vaginal apparatus

HUMAN MALE REPRODUCTIVE SYSTEM

- It is that system which is concerned with sperm formation.
- It is located in the pelvis region. It is formed a pair of testes (also called testicles) along with their accessory ducts (i.e. rete testis vasa efferentia, epididymis and vas deferens), **gland** and the **external genitalia**.

**Diagrammatic sectional view of male pelvis showing reproductive system**

In human beings, male reproductive system is classified into two sex organs. They are

- Essential sex organs (necessary sex organs)
- Accessory sex organs

(a) Essential sex organs: - It includes a pair of testes and is very much required for reproduction (they produce sperms)

(b) Accessory sex organs: - It includes paired and unpaired sex organs.

(i) Paired sex organs or glands

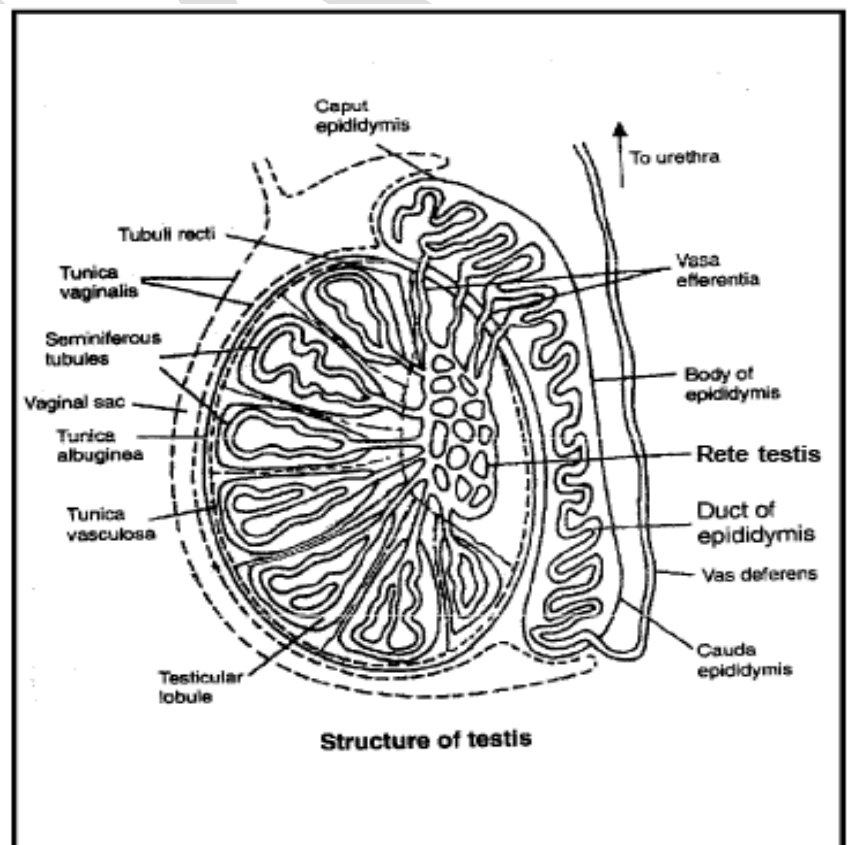
- * Epididymis
- * Vas deferens (male genital duct)
- * Seminal vesicle
- * Ejaculatory duct
- * Cowper's gland (Bulbo urethral gland)

(ii) Unpaired sex organs or gland

- * Scrotal sac
- * Prostate gland
- * Urethra
- * Penis

TESTES

- It is a pair of necessary sex organs located in a highly specialized muscular sac like structure called **Scrotum or scrotal sacs**, present below the abdomen and hanging between the two legs. Scrotal sac is filled tissue fluid called **hydrocoel**.
- They are small sized (4.5 cm x 2.5 x 3 cm), oval shaped, pinkish coloured primary sex organs of male.
- Testis is held in position in scrotum by a small, thick, white fibrous **Gubernaculum** and **spermatic cord**.
- The cavity of scrotal sac is called vaginal coelom and is connected to abdominal cavity. The connection between abdominal cavity and scrotum is called **inguinal canal**.



- Scrotum acts as **thermo regulators** and reduces temperature by 2 – 2.5°C lower than body temperature that is required for spermatogenesis. Thermoregulation is maintained by spermatic cord and smooth muscles of scrotum.
- The body temperature is about 37° C/95° F and a scrotum temperature is 35° C.
- This temperature is ideal for spermatogenesis and surviving of sperms.
- Each testis is covered by three protective layers
 - (i) An outer peritoneal layer called **tunica vaginalis** gives protection.
 - (ii) Middle white, fibrous capsular layer called **tunica albuginea**.
 - (iii) Inner vascular layer called **tunica vasculosa** made up of loose connective tissues associated with blood capillaries, nerve fibers and lymphatic vessels.

- The inner tunica vasculosa shows inward projections and divide the testis into number of compartments called **septa**.
- Each septum consists of number of units called testicular lobules (200-250). Each testicular lobule consists of 2-4 highly convoluted fine tube like structures called **seminiferous tubules**.
- Seminiferous tubules are considered as the structural and functional units of testis.
- Each seminiferous tubules leads to a fine tube called **tubular recti**.
- Each tubular rectum combine to form a complex network called **rete testis**.
- Rete testis later combine to form fine muscular coating tubules called **Vas efferentia**. Each Vas efferentia is lined by cilia and exhibit peristalsis movement and transfer sperm from testis to **epididymis**.

(a) Accessory or Secondary sex Glands

(i) Paired sex glands

(a) Epididymis: -

- On antero-lateral side of each testis an elongated and flattened structure is present called **epididymis**. It is formed of a long (about 6 meters), highly coiled tube.
- Structurally it consists of three parts
 - (1) **Head/caput epididymis/globus major** - It is the first part and swollen part present on cranial (head) side of the testis. It receives sperm cells from vasa efferentia.
 - (2) **Body/carpus epididymis/globus normal**: - It is the second part of the epididymis and is the longest part and lies on lateral side of testis and stores the sperms temporarily.
 - (3) **Tail/cauda epididymis/globus minor**: - It is the terminal part of epididymis and connected to vas deferens.

Functions:

- It carries sperm cells from vasa efferentia to vas deference
- It stores sperm cells for two to three days
- It initiates capacitation of sperms.
- Physiological and functional maturation of the sperms takes place inside epididymis
- During sexual intercourse epididymis constricts and forcibly transfers sperm cells into vas deferens.

(b) Vas deferens/Male genital duct

- A pair of long narrow muscular coated ducts arises from lower part of the epididymis, ascend towards urinary bladder connected to ampulla region of ejaculatory duct is called **vas deference**.
- It measures about **30 cm long**, narrow, muscular and tubular structure.
- Vas deference enters in abdominal cavity through **inguinal** canal and takes a turn over urinary bladder and joins with seminal duct to form ejaculatory duct.

Function:

- Due to muscular coating it exhibits peristalsis as a result sperm cells are forcibly transferring to ejaculatory duct.

Differences between

Vas efferentia	Vas deferens
It is present inside the testis	It is present outside the testis
There are 5-6 pairs of vasa efferentia	Only one pair of vasa deferens
Internal lining of vasa efferentia is made up of cilia	Internal lining of vas deferens is not made up of cilia
It is not covered by muscular coat and fails to exhibit peristalsis	It is covered by muscular coat and exhibit peristalsis
It transfers sperm cells from testis to epididymis	It transfers sperm cells from epididymis to ejaculatory duct

(c) Seminal vesicle

- A pair of slightly elongated (5 cm), muscular and sacculated glands present just below the urinary bladder and attached to ampulla region of the ejaculatory duct.
- Seminal vesicles are involved in production of fluid called **seminal fluid**.
- Seminal fluid contains fructose, prostaglandins, calcium and certain enzymes.

Functions: -

- 70% of semen is secreted from seminal vesicle.
- It secretes alkaline chemicals like **citrate phosphoryl** helps in neutralizing acidic urine in urethra.
- It secretes special chemical called **prostaglandins** which induce contraction of female genital tract for easy sperm entry and help in fusion of gametes.
- Fructose provides energy for the movement of sperms.

(d) Ejaculatory duct: - A pair of ejaculatory duct arises from vas deferens and connected to urethra. The main function is to control the flow of sperm cells and semen into the urethra.

(e) Cowper's gland (Bulbo-urethral gland)

- A pair of white, rounded glands present on either side of urethra is called **Cowper's gland**.
- They are also called **Bulbourethral glands** and secrete mucus.

Functions

- It secretes excess mucus which helps in frictionless movement of penis during sexual intercourse.
- It secretes bicarbonate which neutralizes acidic urine in urethra and also in external orifice of vagina.

(ii) Unpaired sex organs

(a) Scrotal sac: - It is a highly muscular sac like structure present below the abdomen with a pair of testes. In mammals, testes are descended to maintain proper temperature or sperm formation (always 2-3° low temperature).

Note: - Cryptorchidism- Undescended testes/failure of testes to descend into the scrotal sacs/testes remains in the abdomen. This leads to sterility.

(b) Prostate gland

- A single pear-shaped prostate gland present around the neck region of urethra.
- It is involved in production of prostatic fluid.
- The secretion of prostate gland contains citric acid and certain enzymes such as **fibrogenase (fibrinolysis), protease** etc.

Functions

- It secretes alkaline chemicals like citrate and mucus helps to neutralize acidic urine in urethra.
- It secretes 30-40% of semen.
- Prostatic fluid acts as a thinner. The enzymes present in this fluid prevent the coagulation of semen.
- It increases the fluidity of semen and helps in sperm movement.

Note: - In aged male individuals prostate glands expand leads to difficulty in urination (above 50 years). Such individuals undergo prostatectomy (surgical removal of prostate gland).

(c) Urethra

- It is a narrow duct arises from the urinary bladder, pass through the penis and ends in glans penis region.
- It passes through penis and is differentiated into prostatic urethra (2.5 cm), membranous urethra (2.5 cm) and penile urethra (15.0 cm).

Function – Conduction of sperms, secretions of accessory glands and urine.

(d) Penis/External genitalia

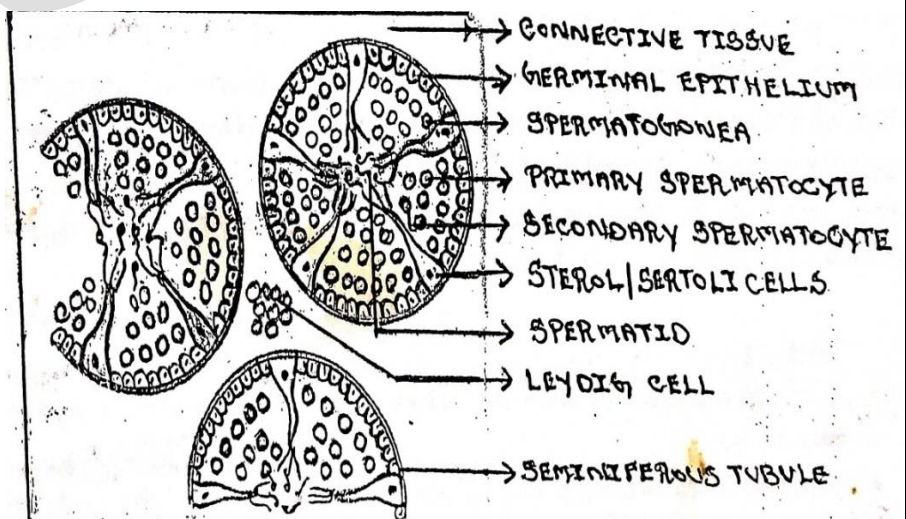
- It is a male copulatory organ present below the abdomen and hanging between the two legs.
- It is made up of erectile tissue called **Cavernosa tissue**.
- The cavernosa tissue presents in the form of three bands or cords.
- The upper or dorsal cords are called **carpus cavernosa** or **corpora cavernosa**.
- The lower or ventral cords are called **Carpus spongiosum**.
- The three cords are bound by connective tissue called **Tunica albuginea** which contains arteries and veins. *Increase in blood pressure in these arteries brings about erection in external genitalia or penis.
- The swollen tip of the penis is called Glans penis and is covered by skin called **foreskin** or **prepuce**.

Function

- The main function is urine discharge and semen discharge during sexual intercourse (Insemination).

T.S of Testis

- T.S of a mature testis consists of three protective layers.
- An outer peritoneal layer called **tunica vaginalis** which gives protection. Middle white fibrous and capsular layer called **tunica albuginea** which provide protection from external and internal vascular layer called **tunica vasculosa** made up of blood capillaries.



- The connective tissue present between the seminiferous tubules contains endocrine cells called **Leydig cells** or **interstitial cells**.
- The Leydig cells are involved in production of a sex hormone called testosterone.
- Each seminiferous tubule is internally lined by cuboidal epithelium called **germinal epithelium**.
- Between cuboidal cells there are elongated cells called Sertoli cells are present.
- Sertoli cells are also called **Sustentacular cells**. These provide nutrition to developing sperms. Hence these cells are also called **Nurse Cells**.
- Seminiferous tubules show the different stages of cells of spermatogenesis.
- Those cells are **spermatogonia, primary spermatocyte, secondary spermatocyte, spermatids, and sperms**.
- The sperms found attached to Sertoli cells, after complete maturation these sperms detach from Sertoli cells. This process is called **spermiation** or the process of release of sperms from the seminiferous tubules is called **spermiation**.

Hormones of Male Reproductive System

- After reaching puberty in males the hypothalamus of brain produces **Gonadotropic Releasing hormone (GnRH)**.
- **GnRH** stimulates pituitary gland to produce gonadotropic hormone.
- There are two types of gonadotropic hormones. They are
 - a) **Follicle stimulating hormone (FSH)** - It stimulates Sertoli cells to produce androgen binding protein and hormone called **Inhibin**.
- Androgen binding protein (ABP) – It binds to testosterone when testosterone concentration increases spermatogenesis takes place.
- **Inhibin** is a hormone that inhibits the production of FSH and LH from pituitary gland.

b) **Luteinizing hormone (LH)** – In males LH is called **interstitial cell stimulating hormone (ICSH)**.

- It stimulates Leydig cells or interstitial cells present between seminiferous tubules to produce a hormone called testosterone. It is a type of androgen.

Functions of Testosterone

- It brings about spermatogenesis
- It is responsible for secondary sexual characters in males such as growth of hair on face, change of voice, development of muscles etc.

Note- The changes that take place on the body after reaching puberty other than reproductive system are together called **secondary sexual characters**.

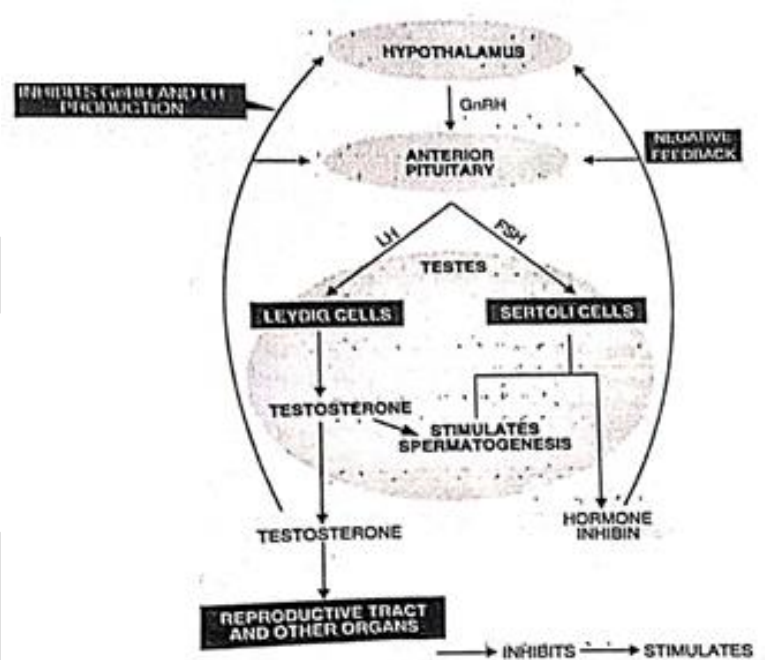


Fig. 3.7. Hormonal control in male reproductive system

Additional points for competitive examination**Male Reproductive System**

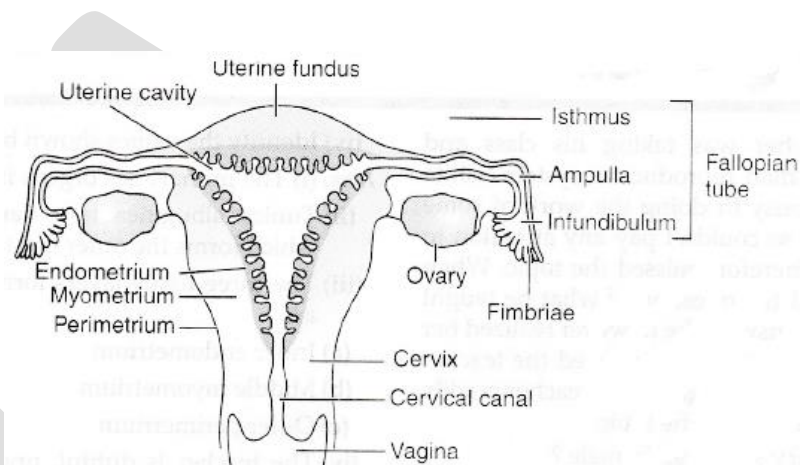
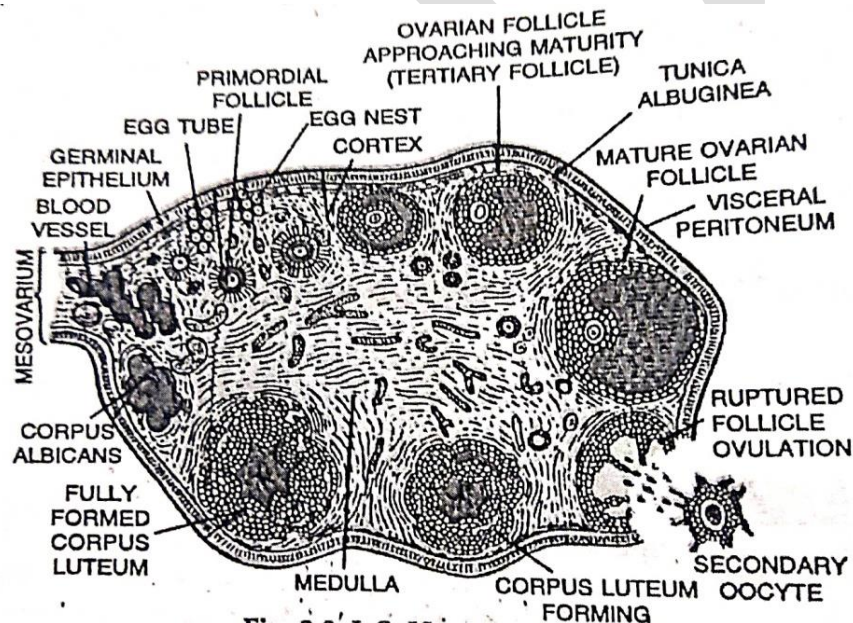
- Testes are also called spermaries.
- In most of mammals, testes are extra – abdominal and lie in scrotal sacs but testes are intra – abdominal in elephants, whales, seals, echidna etc.
- Testes of human males descend in the scrotal sacs during seventh month of development under the stimulating of FSH.
- **Bidders canal** – Male genital duct carrying only sperms and present inside the kidney of frog.
- **Perineal glands** – Sex – attractant secreting glands present in both sexes of rabbit but absent in human beings.
- **Gynandromorph** – Found in *Drosophila*. It has male characters in one half of body and female characters in other half of body.

FEMALE REPRODUCTIVE SYSTEM

- In human beings, female reproductive system can be studied under following headings

(i) **Paired sex organs:** - A pair of ovaries and a pair of fallopian tubes or oviducts.

(ii) **Unpaired sex organs:** -Vaginal orifices (female genitalia or vulva) and uterus.

**Ovary**

- A pair of almond shaped ovaries present on either side of the uterus, attached to inner abdominal wall called **mesovarium** and also attached to uterus through ovarian ligament.
- Each ovary measures about 3 cm length, 2.5 cm width and 1 cm in thickness.
- Transverse section of ovary shows three protective coverings called **outer vesicular peritoneum**, **middle germinal epithelial layer** and **inner tunica albuginea**.
- Entire ovary is filled with a loose connective tissue proper called **stroma/matrix**.

- Matrix consists of two parts. Namely central medulla having more vasculature and peripheral cortex where development of graafian follicle takes place.
- After complete maturation of graafian follicle i.e. the secondary oocyte discharged from ovary into fallopian tube.
- After ovulation the follicular cells produce a yellow colored mass of cells called **corpus luteum**.
- If there is no fertilization, corpus luteum gives rise to **corpus albicans**, which is later degenerated at the end of menstrual flow.

Fallopian tube

- A pair of fallopian tubes arises from uterine part measures about 10-12 cms in length.
- It is made up of highly muscular coating and inner ciliated epithelial tissues.
- It exhibits peristalsis mechanism for the movement of zygote and ciliary action helps in movement of sperm for fertilization.
- Fallopian tube consists of following parts.
 - a. **Fimbriae**-These are small finger like projections present at the tip of fallopian tube having a small opening called **ostium** through which egg enters into fallopian tube.
 - b. **Infundibulum** - It is a funnel shaped structure and also called **oviducal funnel**.
 - c. **Ampulla**-Where fertilization takes place.
 - d. **Isthmus**- It is middle, narrow and ciliated attached to uterine part.

Vaginal orifice

- It is also called female genitalia present just above the anus in between the two legs in the lower part of the abdomen.
- It consists of labia minora, labia majora, vaginal orifice and Bartholin's gland.

Uterus

- It is also called womb in which development of foetus takes place.
- It is highly specialized muscular bag like structure present just below the urinary bladder in the pelvic girdle.
- It measured about 4 cm width, 5 cm length and 2.5 cm thickness.
- Uterus consists of the following parts;

a. Cervix

- It is the lower part of the uterus present just above the vagina is called **cervix**.
- It includes internal orifice and external orifice helps in allowing sperms during sexual intercourse into the uterus and helps in parturition of baby.

b. Vagina

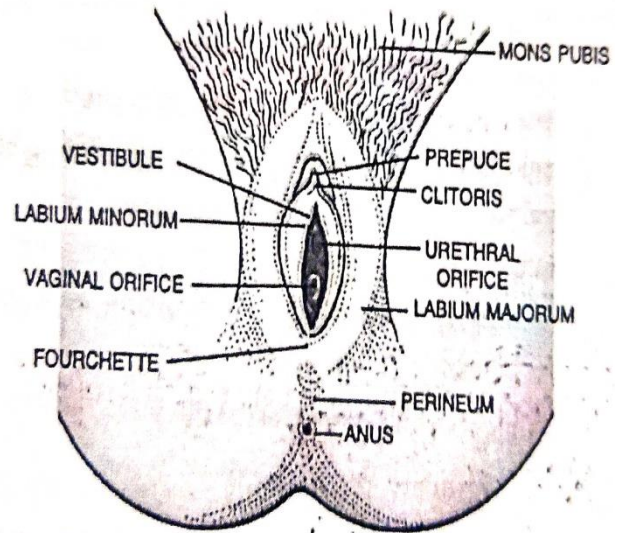
- It is a long (7.5 cm), fibro – muscular tube. It extends backward in front of rectum and anal canal from cervix to the vestibule.
- It is highly vascular tube internally lined by mucous membrane and is raised into transverse folds called **vaginal rugae**. It finally opens in the vestibule by **vaginal orifice**.

Function

- Vagina acts as copulation canal as it receives the sperms from penis during copulation.
- It also acts as birth canal (Cervical canal + vagina) during parturition.
- c. **Uterus body**- It is covered by three protective layers called **outer epimetrium**, **middle myometrium** (made of smooth muscles) and **inner endometrium** (where implantation takes place).
- d. **Fundus**-Anterior part of the uterus which extend on two sides to form uterine part to which fallopian tube attached.

Female External genitalia

- The parts of female reproductive system involved in sexual intercourse or coitus are together called external genitalia.
- External genitalia of female reproductive system include Mons pubis, clitoris, labia majora or upper lip, labia minora or lower lip, vagina and hymen.
- Mons pubis**- It is a cushion of adipose tissue covered by skin and pubic hair. It forms the anterior part of external genitalia.
- Labia majora**- These are two folds of tissues extent from mons pubis and surrounds urethral opening and vaginal opening. **It is homologous to scrotum of male reproductive system.**
- Labia minora**- These are two folds of tissues which are smaller, present next to or inner to labia majora.
- Clitoris**- It is a small fingerlike structure present at the upper junction of two labia minora above urethral opening. **It is homologous to penis of male reproductive system.**
- Vagina and hymen**- Vagina is the opening of female reproductive system. It is covered by thin membrane called **hymen**/virgin knot. It is often ruptured or torn during first sexual intercourse. It can also break by a sudden fall or jolt, insertion of tampon, active participation in some sports like horseback riding, cycling, etc. In some women the hymen may persists even after coitus. So, the presence or absence of hymen is not a reliable indicator of virginity or sexual experience.



Accessory sex glands of female reproductive system

- There are two accessory sex glands associated with female reproductive system. They are
- Bartholin's glands**
- Breast**

(i) Bartholin's glands

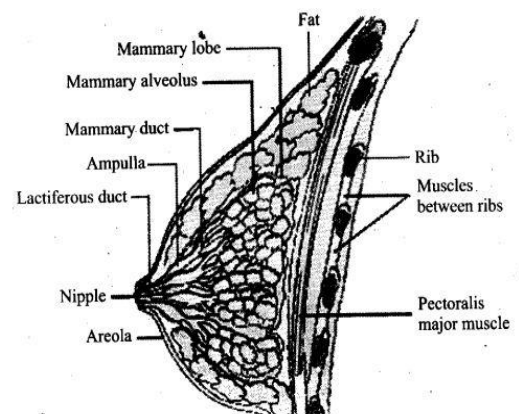
- A pair of rounded glands present behind labia minora of external genitalia is called Bartholin's gland.
- Maturation of Bartholin's glands takes place under the influence of estrogen.
- Bartholin's gland produces excess slimy substance called **mucus**.

Functions

- Mucus helps in frictionless movement of penis during sexual intercourse
- Mucus acts as a lubricant in which sperm cells easily enter into vaginal orifice.
- Mucus always provides an alkaline medium around the vaginal orifice.

(ii) Breast

- A functional mammary gland is characteristic of all female mammals.
- The mammary glands are paired structures (breasts) that contain glandular tissue and variable amount of fat.
- The glandular tissue of each breast is divided into 15–20 mammary lobes containing clusters of cells called **alveoli**.



- The cells of alveoli secrete milk, which is stored in the cavities (lumens) of **alveoli**.
- The alveoli open into **mammary tubules**.
- The tubules of each lobe join to form a **mammary duct**.
- Several mammary ducts join to form a wider mammary ampulla which is connected to **lactiferous duct** through which milk is sucked out.

Additional points for competitive examination

Female Reproductive System

- Female accessory duct system is developed from **Mullerian duct**.
- Mesosalpinx – Mesentery suspending the fallopian tube.
- Mesometrium – Mesentery suspending the uterus.
- Cervix of uterus is formed of most **powerful smooth muscles** in the body.
- Vestibule acts as a **urinogenital sinus**.
- **Perineum** – Area between the fourchette and anus.
- **Bartholin's or Bulbo** – **vestibular glands** of female homologous to **Cowper's glands** of male.
- **Hypermastia** – More than normal number of breasts.
- **Gynaecology** – Study of abnormalities of genital and urinary tract.
- After parturition, a female produces 1.5-2 liters of milk per day.
- In non-pregnant women, mammary glands are scanty.
- After menopause. Mammary glands exhibit **Atrophy** (dead).
- In male individuals, mammary glands are vestigial (non-functional).

GAMETOGENESIS

- **Formation** of haploid gametes from diploid germ cells by meiosis that takes place in gonads is called **gametogenesis**.
- There are 2 types of gametogenesis.
 - (i) Spermatogenesis
 - (ii) Oogenesis

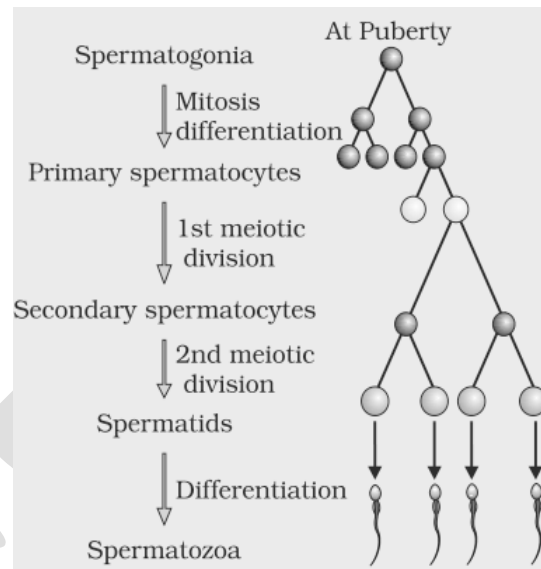
SPERMATOGENESIS

- **The** process of formation of haploid sperms from diploid germ cells by meiosis is called **spermatogenesis**.
- The **process** of formation of haploid sperms from diploid germ cells present in seminiferous tubules by meiosis is called **spermatogenesis**.
- **Spermatogenesis** takes place in seminiferous tubules of the testes.
- The **detailed** process of spermatogenesis can be studied under following steps.
 - (i) Formation of Spermatids
 - (ii) Spermiogenesis/spermioteliolysis
- **(i) Formation of spermatids:** - It takes place in 3 phases
 - (a) Multiplication phase:** - It is the first phase in which diploid germ cells undergo continuous mitotic division and produce large number of diploid cells. These diploid cells produced during multiplication phase are called **spermatogonia**.
 - (b) Growth phase:** - It is the second phase in which spermatogonia are converted into **diploid primary spermatocyte**. During this process, volume of the cell increases due to the formation of more Golgi bodies, mitochondria, microfibrils and capsular proteins.
 - (c) Maturation phase:** - It is the third phase in which the primary spermatocyte exhibit meiosis. During meiosis-I the primary spermatocyte produces 2 equal sized haploid cells called **secondary spermatocyte**. During meiosis-II the secondary spermatocyte produce 4 immature and non-motile male gametes called **spermatids**.

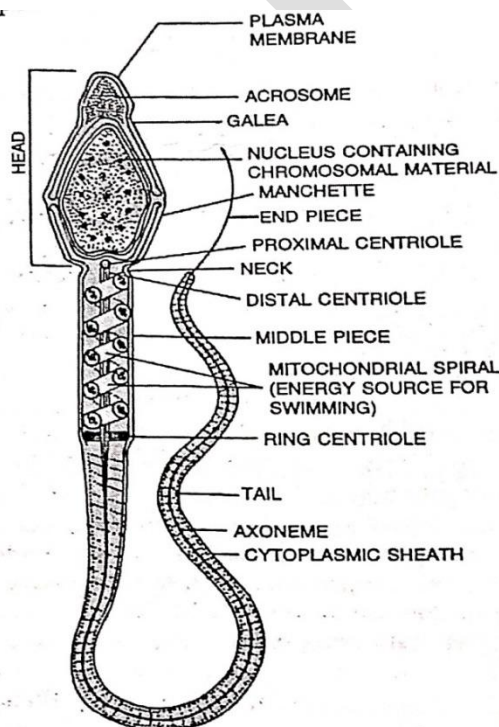
(ii) Spermatogenesis/ Spermioteliiosis

- The **process** of conversion of non-motile, immature spermatids into motile mature sperms is called **spermiogenesis** or **Spermioteliiosis**.
- During spermiogenesis, following morphogenic changes takes place
 - All Golgi bodies fused to form acrosome
 - All mitochondria coiled to form nebenkern
 - All microfibrils are fused to form axial filament/axoneme
 - Proximal and distal centrioles are arranged perpendicular to each other
 - Capsular proteins help in the formation of galea and manchette

Schematic representation of Spermatogenesis



STRUCTURE OF MATURE HUMAN SPERM



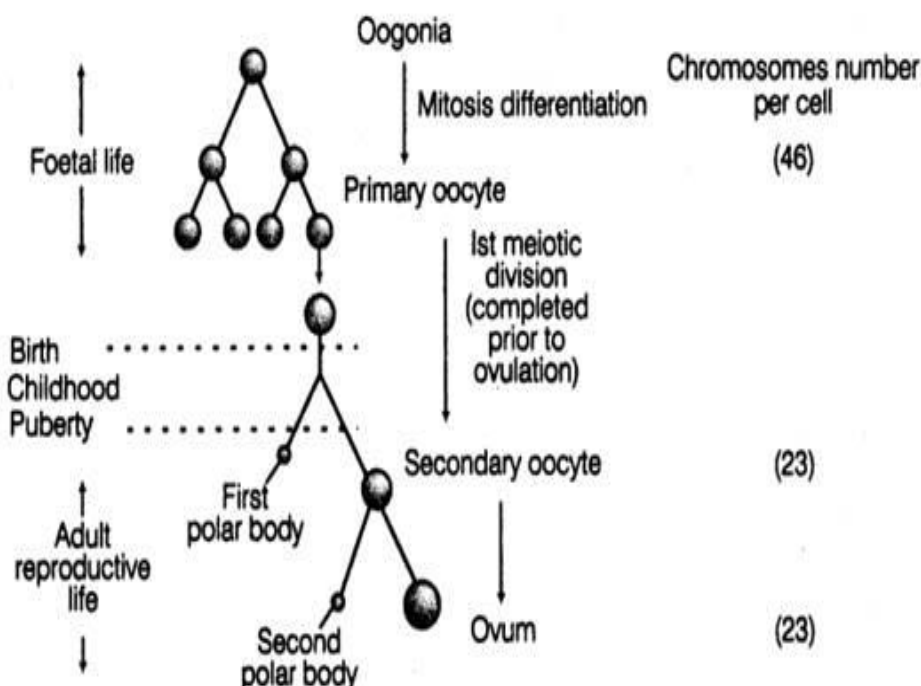
- The sperm is also called as **spermatozoan**.
- It is a small, tad pole, motile, functional, haploid male gamete produced in the testis by spermatogenesis.

- The human sperm measures about 45-55 microns in length.
- Structurally, the sperm is divisible into head, neck, middle piece and tail.
- Head is a small knob like structure present in the anterior end of sperm. It is covered over by a plasma membrane. It encloses acrosome and nucleus.
- Acrosome is a small cap like structure present at the tip of the sperm. It is derived from Golgi complex of the spermatid. It secretes a strong proteolytic enzyme called **hyaluronidase** and **acrosine** helps in the penetration of sperm into the egg during fertilization.
- The nucleus of sperm is called male pro- nucleus. It represents haploid paternal set of chromosomes in the child.
- Neck is the smallest region present between head and tail. It consists of two centrioles. The centriole present towards the nucleus is called **Proximal centriole** and that present towards tail is called **Distal centriole**. Proximal centriole initiates cleavage in zygote.
- The tail is further differentiated into middle piece, main piece and end piece.
- Middle piece consists of mitochondrial spiral, cytoplasmic sheath and axial filament. The main piece consists of cytoplasm sheath and axial filament. End piece consists of only axial filament.
- **Mitochondrial spiral (Nebenkern)** is the spirally coiled mitochondria present in the middle piece and it supplies energy for the movement of sperm and hence middle piece is considered as “**Engine room of sperm**”.
- **Cytoplasmic sheath (Manchette)** is a thick mass of cytoplasm present in the sperm.
- Axial filament (Axoneme) is a thin, long, thread like structure derived from distal centriole. It runs throughout the length of tail. It helps in the locomotion of the sperm.

NOTE: -

- Anterior part of the head is covered by a capsular protein called **Galea**.
- Manchette is the part of the middle piece and is covered by a cytoplasmic protein sheath called **manchette**.

OOGENESIS



Oogenesis in human female showing formation of ovum

- The process of formation of haploid egg from diploid germ cells by meiosis is called **oogenesis** or Formation of ova from diploid germ cell by meiosis that takes place in ovary is called **oogenesis**.
- Oogenesis takes place in the cortex region of ovary.
- The detailed process of autogenesis can be studied by three phases. They are

(i) Multiplication phase: -

- During this phase the diploid primordial germ cell present in ovary undergo continuous mitotic division and produce large number of diploid cells.
- These diploid cells formed after multiplication phase are called **oogonia**.
- Oogonia are produced before birth in foetus.
- New oogonia are not produced after birth. At the time of birth ovary contains millions of oogonia.

(ii) Growth phase: -

- During this phase one of the oogonia increases in size by increasing the volume of nucleus and cytoplasm.
- The enlarged diploid cell formed after growth phase is called **primary oocyte**.
- Primary oocyte is surrounded by granulosa cells to form ovarian follicles.
- Growth phase in oogenesis is the longest phase. It takes place from birth to puberty.
- **At the time of puberty ovary contains 60,000 to 80,000 primary oocytes.**
- Before puberty these primary oocyte degenerate. It is called **follicular atresia. (IMP)**

(iii) Maturation phase: - During this phase the primary oocyte enters into meiosis. During meiosis-I the primary oocyte divides into two unequal sized haploid cells. The larger cell is called **secondary oocyte** and smaller one is called first polar body. Both of these undergo meiosis-II and they give rise to four unequal haploid daughter cells. The larger cell is called **egg/ova/ootid** and the other three smaller ones are called **secondary polar bodies**.

STRUCTURE OF A MATURE HUMAN Egg

- The egg is also called **ovum**. It is a large, spherical, non-motile, functional, inactive, haploid female gamete produced in the ovary by oogenesis.
- The human egg measures about 700-1000 microns in diameter.
- The egg is covered a living plasma membrane called **oolemma**. Which is in turn surrounded by Zona pellucida.
- Zona pellucida is covered by corona radiata. It is cellular layer and secondary egg membrane.
- Zona pellucida is non-cellular layer and primary egg membrane.
- Oolemma encloses cytoplasm called **ooplasm**. Ooplasm contains female haploid nucleus called **germinal vesicle/female pronucleus**.
- Ooplasm also contains protein granules called **cortical granules** which help in fertilization.
- Between the oolemma and zona pellucida two polar bodies (polarocytes) are present.

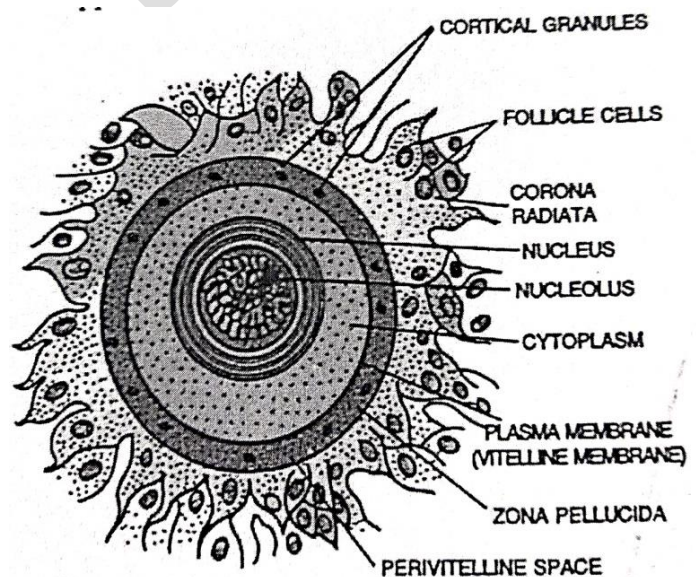


Fig. 3.18. Structure of ovum.

Differences between sperm and ova

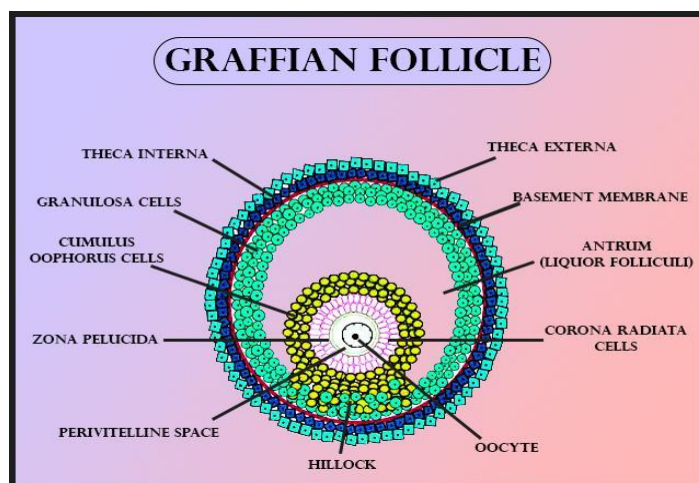
SPERM	OVA
• It is the male gamete	• It is the female gamete
• It is produced in testis	• It is produced in ovary
• It is motile	• It is non-motile
• It is tad pole shaped	• It is spherical shaped
• It is smaller in size	• It is larger in size
• It is active	• It is inactive
• Very small quantity of cytoplasm present	• Very large quantity of cytoplasm present
• Yolk absent	• Yolk present
• Extra membranes are absent	• Extra membranes called egg membranes are present
• Golgi complex are fused to form acrosome	• Golgi complex are diffused
• Mitochondria are fused to form nebenkern	• Mitochondria are diffused

Differences between Spermatogenesis and Oogenesis

Spermatogenesis	Oogenesis
• Formation of sperm	• Formation of ova
• It takes place in testis	• It takes place in ovary
• Growth phase is very short	• Growth phase is very long
• Accumulation of yolk does not take place during growth phase	• Accumulation of yolk takes place during growth phase
• After meiosis-I two equal sized cells are formed	• After meiosis-I two unequal sized cells are formed
• Meiosis-II takes place inside testis	• Meiosis-II takes place outside ovary during penetration of sperm
• After meiosis-II two equal sized cells are formed	• After meiosis-II two unequal sized cells are formed
• The cells formed after meiosis-II are not matured male gametes	• The cells produced after meiosis-II is the mature female gamete called ova
• Four sperms are produced	• Only one ovum produced
• Takes place in two steps	• Takes place in one step

Structure of a mature Graafian follicle

- The mature mammalian ovarian follicle is called graafian follicle.
- A single graafian follicle is matured in one of the ovaries during menstrual cycle.
- It is usually spherical in shape covered by an **outer capsular theca externa** and **inner cellular theca interna**.
- Entire graafian follicle is covered by membrane called **membrana granulosa** is made up of number of follicular cells.
- A cavity is present in the graafian follicle in between the egg and wall of the graafian follicle is called **antrum**. It is filled with a fluid called **liquor folliculi**.



- The egg is attached to the wall of the graafian follicle by a mass of cells called **cumulus oophorus/discus proligerous/germ hill**.

Additional points for competitive examination

Gametogenesis and Gametes

- Spermatogenesis** is a **continuous process**, while **oogenesis** is a **discontinuous process**.
- Spermatogenesis starts after the attainment of puberty, while oogenesis begins even during foetal development.
- In spermatogenesis, spermatogonium produces four sperms while in oogenesis, one oogonium produces one ovum and 2 or 3 polar bodies.
- Human sperm has **clupein protein** on its nucleus.

Types of Eggs

I. On the basis of amount of Yolk.

- Microlecithal or Oligolecithal eggs** – These are small sized eggs with very small amount of yolk. **E.g.:** - Hydra, sea urchin, Amphioxus, Marsupials, Eutherians.
- Mesolecithal eggs** – These contain moderate amount of yolk. **E.g.:** - Annelids, molluscs, Petromyzon, ung fishes, amphibians (frog).
- Macrolecithal or Polylecithal eggs** – These contain very large amount of yolk. **E.g.:** - Insects, reptiles, birds and prototherian mammals.

II. On the basis of distribution of Yolk

- Homolecithal or Isolecithal eggs** – In these eggs, small amount of yolk is uniformly distributed in the cytoplasm. **E.g.:** - Animals with microlecithal eggs.
- Telolecithal eggs** – In these eggs, the moderate amount (**e.g.**, mesolecithal eggs) or large amount (**e.g.**, polylecithal eggs) of yolk concentrated in one hemisphere called **vegetal pole**. Other side of egg is called **animal pole**. It is found in **mesolecithal eggs** (**e.g.**, of frog) and polylecithal and meioithal and highly **telolecithal eggs** (**e.g.**, reptiles, birds and prototherians). In meiolecithal egg, the cytoplasm is confined in a disc – like area at animal pole, called **blastodisc**.
- Centrolecithal eggs** – In these eggs, the yolk is concentrated at the center of the egg surrounded by a peripheral layer of cytoplasm. **E.g.**, in insects.

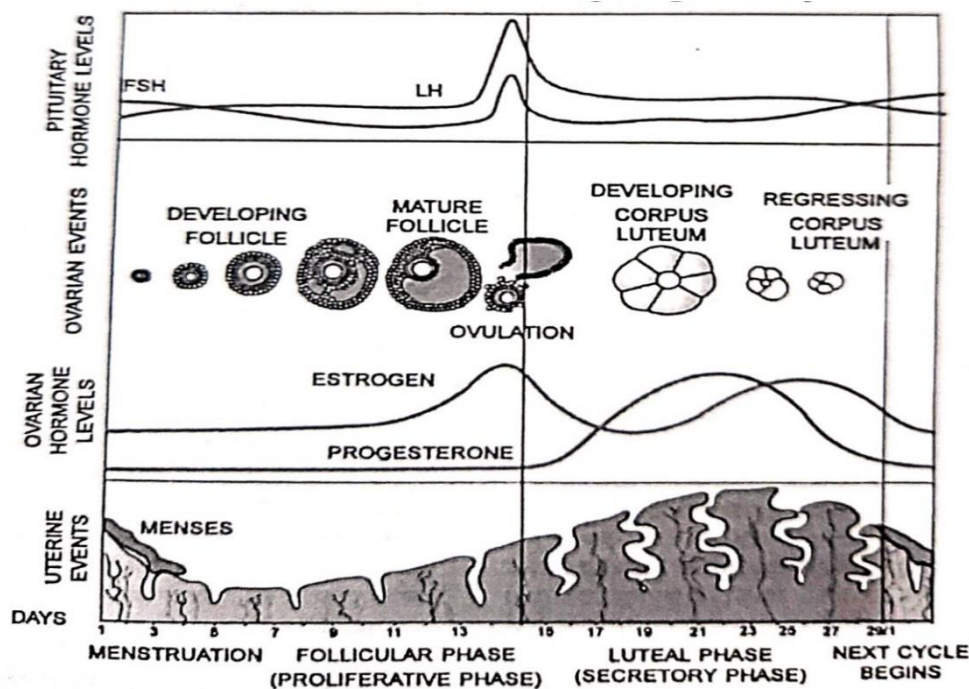
REPRODUCTIVE CYCLES

- A series of rhythmic changes that takes place in female reproductive system after reaching puberty is called **reproductive cycle**.
- In mammals there are two types of reproductive cycles
 - Oestrus cycle
 - Menstrual cycle

Differences between

Oestrus cycle	Menstrual cycle
It occurs in non-primates like cow, sheep, dog, cat	It occurs in primates like human beings, chimpanzee, gorilla
There is heat period	Heat period is absent
Mating should take place in heat period	Mating can occur at any time
Menstrual flow absent	Menstrual flow takes place
Breeding period is very short	There is no specific breeding period
There is no bleeding after detachment of uterine endometrium	Bleeding takes place during menses

Menstrual cycle



- It is a regular and cyclic process in which some complex process occurs in the female reproductive system in every 28 days or the preparation and detachment of endometrium (decidua) in a non-pregnant woman that takes place for every 28 days is called **menstrual cycle**.
- Menstrual cycle takes place in 4 phases
 - (a) Pre-ovulatory phase
 - (b) Ovulatory phase
 - (c) Post ovulatory phase
 - (d) Menstrual phase

a) Pre-ovulatory phase/Proliferative phase

- It takes place from 5th day to 13th day.
- During this phase pituitary gland secretes FSH.
- FSH stimulates ovary to produce ovarian follicle.
- Ovarian follicle is converted into graafian follicle which produces a hormone called **estrogen**.
- Estrogen brings about thickening of endometrium (decidua).

b) Ovulatory phase: -

- It takes place from 14th day in menstrual cycle.
- During this phase pituitary gland secretes LH.
- LH brings about ovulation (releasing of oocyte from graafian follicle into fallopian tube is called **ovulation**).

c) Post ovulatory phase/Secretory phase

- It takes place from 15th to 28th day in menstrual cycle.
- During this phase the empty graafian follicle forms corpus luteum.
- Corpus luteum secretes a hormone called **progesterone**.
- Progesterone brings about further thickening of endometrium and inhibits the production FSH and LH from pituitary gland.
- If the fertilization does not take place corpus luteum degenerates into **corpus albicans**.

- Corpus albicans do not produce estrogen and progesterone.
- In the absence of estrogen and progesterone the prepared endometrium (decidua) detaches and blood vessels rupture which leads to bleeding.

d) Menstrual phase: -

- It takes place from 1st day to 5th day in menstrual cycle.
- During this phase the detached endometrium (decidua) blood and unfertilized egg come out of female reproductive system. This flow is called menstrual flow.
- During this phase the endometrium will be very thin.

Hormones of Female reproductive system

- After reaching puberty in females hypothalamus releases a hormone called **Gonadotropic releasing hormone (GnRH)**.
- GnRH stimulates pituitary gland to produce gonadotropic hormone. There are two types

(1) Follicle stimulating hormone (FSH)

- It converts ovarian follicle into graafian follicle.
- It stimulates graafian follicle to produce estrogen.

(2) Luteinizing hormone (LH)

- It brings about ovulation.
- It converts empty graafian follicle into corpus luteum.
- It stimulates corpus luteum to produce hormone called progesterone.

Functions of estrogen

- It initiates menstrual cycle.
- It brings about thickening of endometrium (decidua).
- It is responsible for secondary sexual characters in females.
- Some of the secondary sexual characters are growth of pubic hair, enlargement of hip region, development of breast, accumulation of adipose tissue below the skin.

Functions of Progesterone

- It brings about further thickening of endometrium.
- It inhibits the production of FSH and LH from pituitary gland.
- It maintains pregnancy. Hence it is also called **Pregnancy hormone**.

NOTE:

- Pre-ovulatory phase is also called **estrogen phase** and post-ovulatory phase is also called **Progesterone phase**.
- First menstrual cycle that takes place at the age of 11 or 12 years is called **Menarche**.
- The last menstrual cycle that takes place at the age of 45 years is called **Menopause**.

Additional points for Competitive Examination

- **Esrogen** plays a central role in the menstrual cycle.
- Level of estrogen is maximum on 12th day of menstrual cycle while level of LH is maximum on 13th day of cycle.
- Ovulation always occurs on 14th days before menses as the luteal phase has fixed duration, while that of follicular phase varies.

- Corpus luteum is fully formed on 19th day of menstrual cycle, and level of progesterone is maximum on 21st day of the cycle.
- During menstrual phase, the thickness of uterine endometrium is just above 1 mm thick.
- There is no permanent climacteric period in human male due to presence of **Type – A spermatogonia**.
- In human female, administration of progesterone 3 or 4 days before the expected date can defer menstruation for a few days.
- The day on which discharge of blood from the vagina begins is taken as the first day of the menstrual cycle.
- Cyclic menstruation is an indicator of normal reproductive phase and extends between menarch and menopause.
- The average age of menopause in India is 48 years. If it occurs before the age of 40, it is called premature menopause.
- Menstrual cycle is absent during pregnancy, while may be suppressed during lactation and is permanently stopped at menopause.
- About 13 mature eggs are released from two ovaries of female **in a year**, so about **416 eggs (13x32 years)** are ovulated from the ovaries during the whole reproductive period of human female.
- Two ovaries alternate in ovulation.
- Menstruation is also called “**Weeping of uterus for the lost ovum**” or “**Funeral of unfertilized egg**”.
- Menstrual cycle occurs only in primates, while most of mammals show oestrous cycle.
- Absence of menstruation in a female after puberty is an indicator of pregnancy.
- Frog is a **spontaneous breeder** and is also a **seasonal breeder**.
- Menstruation may stop due to a number of reasons like establishment of pregnancy, stress, poor health etc.

FERTILIZATION AND IMPLANTATION

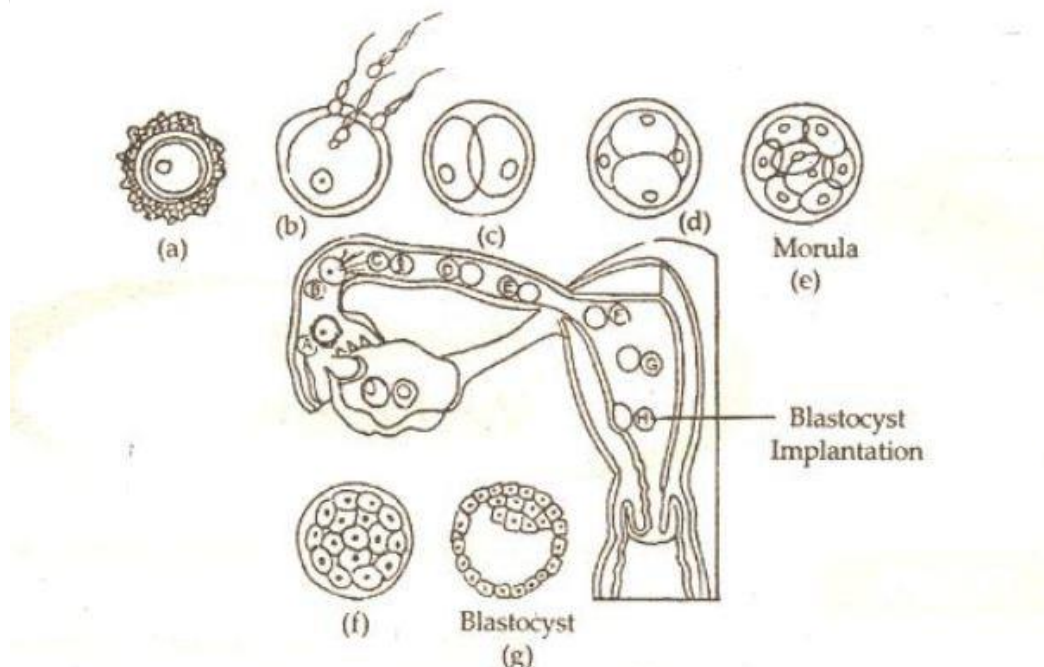


Fig. : Transport of ovum, fertilisation and passage of growing embryo through fallopian tube

- The complete fusion of haploid male gamete and haploid female gamete to form diploid zygote is called **fertilization**.
- In human beings, fertilization is said to be internal (takes place in the female genital tract) and cross fertilization.

- Fertilization takes place in the ampullar region of fallopian tube.
- Fertilization always takes place within two days after ovulation
- The zygote moves in fallopian tube to reach uterine cavity.
- When the zygote moves in fallopian tube mitotic division which is not associated with growth takes place.
- Such mitotic division which is not associated with growth that takes place in zygote is called **Cleavage**.
- Cleavage results in formation of 2 cells, 4 cells, 8 cells and 16 cells.
- The cells formed after cleavage are called **blastomeres**.
- Cleavage results in formation of a mass of cells called **morula**. It is formed on 5th day after fertilization.
- On 6th day some cells in morula become flat and form outer layer called **Trophoblast**. Some cells remain spherical and are found attached to trophoblast in one region. These cells are together called **inner cell mass** or **embryoblast**.
- The trophoblast encloses a cavity called **blastocoel**.
- This single layer hollow structure formed from morula is called **Blastocyst (Blastula)**.
- The blastocyst remains in uterine cavity for about 24 hours. It obtains nutrients from uterine fluid.
- On 7th day the blastocyst attaches to the wall of uterine cavity (endometrium). This process is called **Implantation** or **nidation**.
- Implantation takes place to get nutrients from the uterine wall.

Additional points for Competitive Examination

- **Cleavage** is the **first phase** of embryonic development.
- Cleavage divisions are the slowest in the mammals, including humans.
- In humans, cleavage divisions are **asynchronous** from the very beginning.
- **Compaction** is the process by which morula produces two types of blastomeres.
- Estrogen and Progesterone are called **synergetic hormones** as both act on endometrium and help in implantation.

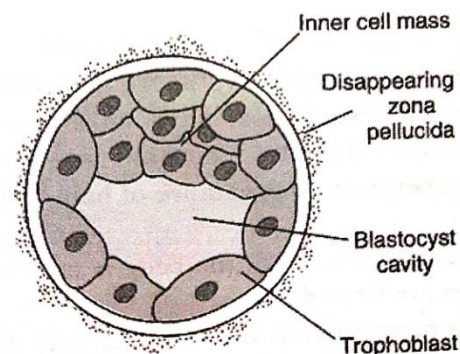
Types of Cleavage

(1) On the basis of extent of division of zygote

- Holoblastic cleavage** – In this, the zygote divides completely, so the number of blastomeres goes doubling with each division. It is again of two types.
 - Holoblastic equal cleavage** – The blastomeres are of the same size. It is found in alecithal and microlecithal eggs. **e.g.**, in Starfish, Herdmania, Amphioxus, Marsupials etc.
 - Holoblastic unequal cleavage** – The blastomeres are of two types - smaller micromeres and larger megameres. It is found in mesolecithal eggs. **e.g.**, Frog.
- Meroblastic cleavage** – In this, cleavage passes only through the yolk free cytoplasmic part of the zygote while the yolk part of zygote remains uncleaved. It is found in **polylecithal** and **meiolecithal eggs** (Meroblastic and discoidal cleavage) of reptiles, birds and prototherians and **centrolecithal eggs** (meroblastic and superficial cleavage) of insects.
 - Ova fertilized by single sperm are called **monospermy**.
 - Ova penetrated by a greater number of sperms are called **polyspermy**.

Structure of Blastocyst

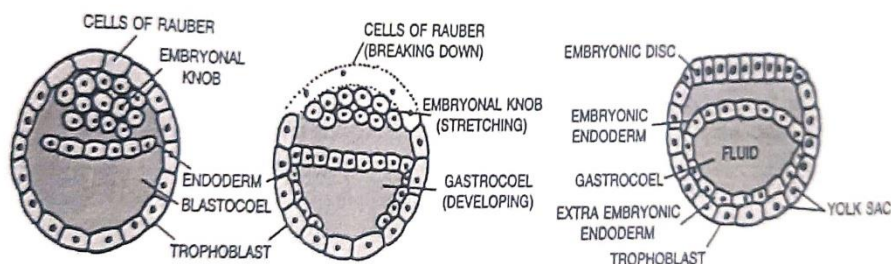
- Blastula of human beings is called **Blastocyst**.
- It is single layered hollow structure formed from morula.
- It is still covered by **zona pellucida**. It encloses a cavity called blastocoel.
- It contains outer single layered flat cells called **Trophoblast**.
- A group of spherical cells found attached to trophoblast and project in blastocoel are together called **inner cell mass** or **embryoblast**.
- Trophoblast develops into extra embryonic membranes and inner cell mass develop into embryo.



IMPLANTATION/CNIDATION

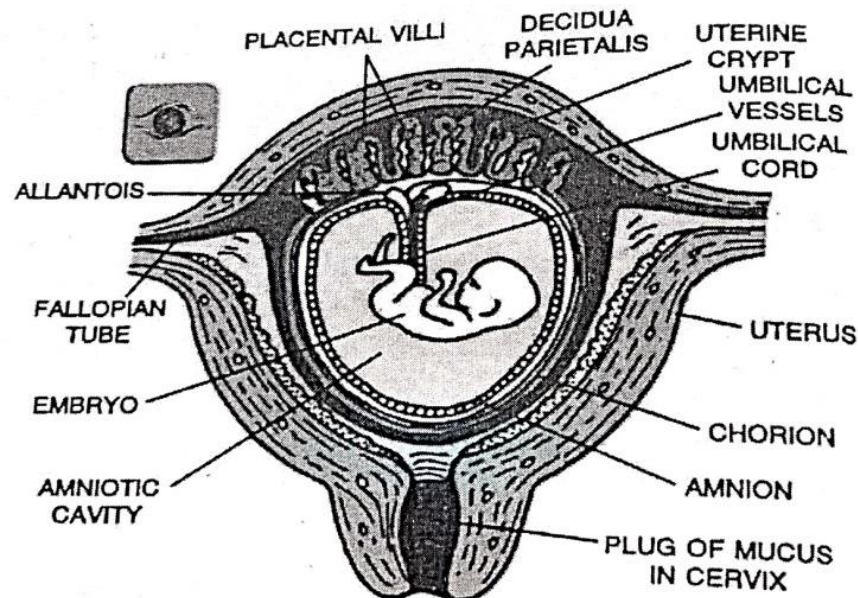
- The process by which the blastocyst is attached into the wall of uterine cavity or endometrium is called **implantation** or **cnidation**.
- It takes place on 9th day after fertilization. It is influenced by a hormone called **progesterone**.
- During implantation the trophoblast splits into two layers. The outer layer is called **synchititrophoblast** and inner layer is called **cytotrophoblast**.
- Synchititrophoblast releases lytic enzymes on endometrium. The enzymes degenerate the cells of endometrium due to which a cavity is formed.
- The blastocyst enters into this cavity and the endometrium regenerate over blastocyst.
- During implantation the endometrium becomes soft and highly vascular. This soft and highly vascular endometrium is called **decidua**.
- After implantation the decidua present above the blastocyst is called **decidua capsularis**.
- The decidua present below the blastocyst is called **decidua basalis**.
- The decidua present opposite to implantation is called **decidua parietlis**.

Formation of three germ layers



- After implantation, the inner mass cells of the blastocyst undergo repeated divisions and reorganized into three principle layers called ectoderm, mesoderm and endoderm.
- After implantation the cells found towards the periphery of embryo blast (inner cell mass) of the blastocyst become flat.
- These flat cells spread along with trophoblast and covers the blastocoel.
- These flat cells represent the endoderm; the cells of embryoblast elongate and become columnar. These elongated cells represent embryonic disc.
- The cells of embryonic disc exhibit continuous mitotic division and produce new cells.
- These new cells spread between embryonic disc and endoderm. These cells represent mesoderm and the embryonic disc is called ectoderm.
- Ectoderm, endoderm and mesoderm are the primary germ layers developed from inner cell mass.
- The trophoblast gives rise to extra embryonic membranes. They are amnion, chorion, allantois and yolk sac.

PLACENTA



- After implantation, the blastocyst forms finger like projections called **chorionic villi**.
- The chorionic villi and the uterine tissue together form the structural and functional unit between the developing embryo and maternal body called **placenta**.
- The placenta is connected to the embryo with umbilical cord which helps in the transport of substances to and fro the embryo.
- Placenta also acts as endocrine gland and produce hormone called **HCG [Human Chorionic Gonadotropin]**, **hPL [Human Placenta Lactogen]**, **estrogen**, **progesterone** and **relaxin**.

FUNCTIONS OF PLACENTA

- It helps in transportation of nutrients such as glucose and amino acids present in maternal blood into foetal blood by diffusion.
- It helps in exchange of oxygen and carbon di oxide between maternal blood and foetal blood by respiration.
- It helps in the excretion of nitrogenous wastes produced in foetus enters into maternal blood through placenta.
- Placenta is involved in storing of glycogen and iron. It performs the functions of liver.
- Placenta allows the antibodies present in maternal blood to enter into foetus which provide innate immunity to the newly born baby.
- Placenta acts as impermeable and do not allow the hormones in maternal blood to enter into foetus.
- Placenta is involved in the production of four types hormones and acts as endocrine glands.
- **Human chorionic gonadotrophic hormone (hCG):** - It maintains corpus luteum to maintain continuous production of progesterone.
- **Human placental Lactogen (hPL)** - It activates prolactin which stimulates mammary glands to produce milk.
- **Progesterone:** - Placenta after three months starts producing progesterone to maintain pregnancy.
- **Relaxin:** - It brings about dilation of cervix and helps in parturition.
- Placenta fails to prevent the movement of pathogen into foetus.
- Placenta also fails to prevent the movement of drugs taken during pregnancy into foetus from maternal blood. These drugs affect the development of foetus.

Additional points for Competitive Examination

- Placenta in which there is intimate contact between foetal tissue and maternal tissue is called **deciduate placenta**. In such condition during parturition maternal tissue is damaged and leads to bleeding.
- Placenta fails to prevent the movement of drugs from maternal blood into foetal blood taken during pregnancy. These drugs affect the development of foetus.
- **Thalidomide** is a sedative and pain killer given at the time of labour. This drug causes abnormal development of limbs (seal like limbs) in newly born baby. This condition is called **Phocomelia**.
- Placenta also fails to prevent the movement of pathogens from maternal blood into foetal blood if mother is infected at the time of pregnancy.
- Presence of HCG in urine confirms pregnancy. This test is called **gravindex test**.
- Amnion, chorion and allantois are formed only in amniote embryo as their development occurs on land either inside egg or in the uterus of mother. These are not formed in anamniotes as their development occurs in water so there is no problem of desiccation, supply of oxygen and removal of wastes.
- **Amnion** is the innermost embryonic membrane and never participates in placenta formation.
- **Chorion** is the outermost embryonic membrane and always participates in placenta formation.

Types of placenta**(i) On the basis of foetal membranes involved in the formation of placenta**

Placentae are of 3 types

- (a) **Yolk sac placenta** – Formed of Yolk sac and chorion. **e.g.**, marsupials like kangaroo, opossum.
- (b) **Allantoic placenta** – Formed of allantois and chorion. **e.g.**, most of eutherians
- (c) **Chorionic placenta** – Formed of only chorion. **e.g.**, primates

(ii) On the basis of intimacy between foetal and maternal blood, placenta is of 5 types on the basis of number of layers separating their bloods

- (a) **Epitheliochorial** – 6 layers – **e.g.**, odd hoofed mammals (horse, pig)
- (b) **Syndesmochorial** – 5 layers – **e.g.**, Even – hoofed mammals (Cow, sheep, goat, camel etc.)
- (c) **Endothelialchorial** – 4 layers – **e.g.**, Carnivores (Dog, cat, lion, tiger etc.)
- (d) **Haemochorial** – 3 foetal layers – **e.g.**, Primates (Monkey, apes, lemurs, human female)
- (e) **Haemoendothelial** – only foetal blood capillary – **e.g.**, rodents and lagomorphs (rabbit)

PREGNANCY/GESTATION

- The period taken by the embryo for its development is called **pregnancy** or **parturition**. Gestation period in humans lasts for 9 months (280 days).

ORGANOGENESIS

- The process of formation of organs and organ system during early development is called organogenesis.
- After one month of pregnancy heart is formed and starts functioning.
- In **First month**, there is formation of blood vessels and neural tube. The formation of heart is the first sign of growth.
- Limbs and digits are formed at the end of second month of foetus.
- In the **second month** primitive characters such as tail and pharyngeal gill slits are also formed.
- At the end of **first trimester or third month (12th week)** all primitive characters disappear and external genitalia is developed. (**Sex of the foetus can be determined in the third month**).
- In the **Fourth month** all organ systems develop and the developing foetus will have human appearance.

- At the end of **Fifth month** first movement of foetus takes place.
- At the end of **Sixth month** or **Second trimester (24th weeks)** body is covered by tiny hair (Lanugo), eye lids separate and eye lashes are formed.
- On the **Seventh month** complete development of foetus takes place and on the **Eighth month** if the foetus is male testicles enters into scrotum.
- At the end of ninth month the fully developed foetus exerts pressure of uterine wall which lead to parturition.

PARTURITION

- Expelling of fully formed young ones from the mother's uterus through vaginal orifice after gestation period is called **parturition** or **it is the process of giving birth to a baby**.
- In human being, gestation period is about 280 days/9.5 months from the start of the mother's last menstruation.
- Parturition is induced by a complex neuroendocrine mechanism.
- The fully developed foetus and placenta induces mild contraction of uterus. It is called **foetal ejection reflex**.
- During parturition, foetus secretes hormones from its adrenal gland. These hormones diffuse into mother blood and reaches posterior part of the pituitary gland to produce a special hormone called **oxytocin** (birth hormone).
- Oxytocin stimulates uterine endometrium which helps in uterine contraction. At the same time, oxytocin stimulates placenta to produce **prostaglandin** chemicals, and these chemicals stimulates uterine wall and also internal orifice of the cervix for the easy parturition of baby.
- During parturition time, labour pain develops i.e. forceful contraction of uterine wall leads to more pain during parturition is called **labour pain**.
- As the birth approaches, the foetus fully positioned in such a way that head is facing towards cervix.
- Parturition can be studied under three stages

(a) Dilation stage: -

- It takes about 12 hours; during this stage regular peristaltic contraction of uterine wall begins from its top and moves downwards. As the result, foetus moves downwards to the cervix.
- Here amnion and chorion ruptures, amniotic fluid passes out leads to lubrication of vaginal orifice.
- Now foetal placenta pulled out from the uterine wall and foetus becomes free from maternal tissues.

(b) Expulsion: -

- It takes about 20 minutes.
- Here contraction of uterine wall is more powerful followed by abdominal wall contraction as a result baby's head comes out of the cervix.
- Here, umbilical cord is cut near the baby; this frees the baby totally from mother.
- Now relaxin hormone induces relaxation of pelvic girdle ligament which leads to easy birth.

(c) After-birth

- It takes about 12 minutes.
- After baby's birth, uterus become relaxed as a result, placenta becomes free from the implantation site.

- Now placenta, umbilical cord, foetal membrane ruptured, blood capillaries and same amount of blood expelled out due to series of rhythmic contraction and relaxation of myometrium part of uterus. Now, immediately after birth cervix released to prevent excess bleeding.
- After 25-30 days, uterus returns fully to its non-pregnancy state.

LACTATION

- Synthesis, secretion and ejection of milk in female after child birth are called **lactation** or production of milk in mammary glands during pregnancy and after pregnancy is called **lactation**.
- The milk that is produced for 2 or 3 days after parturition is yellowish in colour and contains large number of antibodies (IgA) is called colostrum or first milk.
- The antibodies present in colostrum provide innate immunity to the newly born baby.
- Mechanism of lactation can be studied under 2 phases

(a) Milk secretion

- Milk production takes place after the formation of placenta under the influence of prolactin hormone.
- But maximum milk secretion takes place after the child birth because the sucking of the nipple around the areola part stimulates receptors of hypothalamus **PRF- Prolactin releasing factor**.

(b) Milk ejection

- It involves ejection of milk from mammary glands and laticiferous ampulla by oxytocin.

COMPOSITION OF MILK

- Milk produced after few days of child birth which is slightly yellowish in colour with excess antibodies is called **Colostrum**.
- It consists mainly the antibody IgA
- Milk consists of Na, K, Ca, P, lacto albumin, casein protein, lipid, lactose sugar, various types of antibodies and excess water.

NOTE

- **Progesterone** is called **pregnancy hormone** and **Oxytocin** is called **birth hormone** or **labor hormone**.
- Oxytocin is usually injected to the pregnant woman at the time of parturition to facilitate contraction of uterine wall.

One Mark Questions:

1. Name the male reproductive structure of the flower.
2. Name the female reproductive structure of the flower.
3. How many thecae are present in a typical mature anther of angiosperms?
4. What is a ditheous anther?
5. How many microsporangia are present in a typical anther of angiosperms?
6. How many microsporangia are present in each lobe of an anther?
7. How many wall layers are present in a microsporangium?
8. Name the innermost wall layer of microsporangium?
9. What is the function of tapetum?
10. What is a sporogenous tissue?
11. Define microsporogenesis.
12. Name the tissue in the microsporangium which produces microspores.
13. Which type of cell division occurs in pollen mother cell during microsporogenesis?

14. If a microsporangium has 450 pollen mother cells, how many microspores are produced?
15. What is the ploidy of functional microspore in flowering plants?
16. What does pollen grain represent?
17. Name the outer layer of pollen grain.
18. Name the layer of pollen grain which is generally made of sporopollenin.
19. Name the organic chemical present in exine of pollen grain.
20. What is the biological significance of sporopollenin?
21. What are germ pores?
22. Name the region on the exine of pollen grains where sporopollenin is absent.
23. Name the organic chemical present in intine of pollen grains.
24. Name the large cell of a mature pollen grain.
25. Name the small cell of a mature pollen grain.
26. Why pollen grains are very well preserved as fossils?
27. How many cells do a mature male gametophyte has in majority of angiosperms just before it germinates on the stigma?
28. Name the cell in the male gametophyte of angiosperms which produces the male gametes.
29. Name the cell in the male gametophyte of angiosperms which produces the pollen tube.
30. Name the solution used to promote pollen germination and formation of pollen tube.
31. Arrange the following terms in the correct developmental sequence:
32. Pollen grain, Sporogenous tissue, Microspore tetrad, Pollen mother cell, Male gametes
33. Give an example for a plant that causes pollen allergy.
34. How are pollen grains preserved?
35. What is a monocarpellary pistil?
36. What is a multicarpellary pistil?
37. What is a syncarpous pistil?
38. What is an apocarpous pistil?
39. What is megasporogenesis?
40. Name the type of cell division that the megaspore mother cell undergoes?
41. What is monosporic development of embryo sac?
42. What is the ploidy of functional megaspore in flowering plants.
43. What is chalaza?
44. Name the basal part of an angiosperm ovule.
45. Name the region of the ovule present opposite to the micropylar end.
46. What is micropyle?
47. Name the opening of the angiosperm ovule where the integument is absent.
48. Name the opening of the ovule through which pollen tube enters.
49. Name the stalk of the angiosperm ovule.
50. What is funicle?
51. Name the region on the body of the ovule to which the funicle is attached.
52. What is hilum?
53. What are the protective envelopes of the ovule called?
54. What are integuments?
55. What is nucellus?
56. Name the nutritive tissue with reserve food present in an angiosperm ovule.
57. Name the mass of cells enclosed within the integuments in an ovule.
58. What is free-nuclear division of functional megaspore?
59. Name the cells present at the micropylar end in the embryo sac.
60. What is 'filiform apparatus'?

61. Name the special cellular thickenings of the synergids.
62. What is the function of filiform apparatus?
63. Mention the number of nuclei found in a typical mature angiosperm embryo sac.
64. Mention the number of cells found in a typical mature angiosperm embryo sac.
65. Name the structure found inside the megasporangium which is 7-celled and 8-nucleated.
66. Define pollination?
67. The distribution of bryophytes and pteridophytes is limited with respect to sexual reproduction. Why?
68. Define autogamy.
69. What are chasmogamous flowers?
70. Give one example for a chasmogamous flower.
71. What are cleistogamous flowers?
72. Give one example for a cleistogamous flower.
73. Why cleistogamous flowers are invariably autogamous?
74. Cleistogamy is advantageous to the plant. Why?
75. Define geitonogamy.
76. Even though geitonogamy is functionally cross pollination involving pollinating agents, it is genetically similar to autogamy. Why?
77. Define xenogamy.
78. Papaya plants exhibit xenogamy only. Why?
79. Name the type of pollination that brings genetically different types of pollen to the stigma.
80. Mention one abiotic agent used by plants to achieve pollination.
81. Mention one biotic agent used by plants to achieve pollination.
82. Give one example for wind pollinated plant.
83. Give one example for water pollinated plant.
84. How are pollen grains protected from wetting in water pollinated species?
85. Which are the dominant pollinating agents among insects?
86. Mention one important feature of insect pollinated flower.
87. Mention one floral reward provided by the flower to a pollinating animal.
88. Give an example for a plant that gives floral reward by providing safe place to lay insect eggs.
89. Why wind and water pollinated flowers produce enormous amount of pollen when compared to the number of ovules available for pollination?
90. What is the floral reward that the flower of *Amorphophallus* provides to pollinating agent?
91. Name the plant which has the tallest flower.
92. Some insects are called 'pollen robbers' or 'nectar robbers'. Why?
93. Why many flowering plants have developed devices to discourage self pollination and encourage cross pollination?
94. Mention one strategy evolved to prevent self-pollination in flowers.
95. What is self incompatibility?
96. What is pollen-pistil interaction?
97. What is artificial hybridization?
98. What is emasculation?
99. Why bisexual flowers of a plant are emasculated in artificial hybridization?
100. Why emasculated flowers are covered with bags of butter paper?
101. What is bagging with reference to artificial hybridisation?
102. What is triple fusion?
103. What is double fertilization?
104. What does the primary endosperm cell develop into?
105. What is the ploidy of the primary endosperm cell?

106. Expand PEN.
107. What is embryogeny?
108. Why do you think that the zygote is dormant for sometime immediately after fertilization in an ovule?
109. What is epicotyl?
110. What is hypocotyl?
111. What is scutellum?
112. What is coleorrhiza?
113. What is coleoptile?
114. What is seed?
115. What are non-albuminous seeds?
116. What are albuminous seeds?
117. Give an example for a plant which produces albuminous seeds.
118. Give an example for a plant which produces non-albuminous seeds.
119. Seeds of pea, groundnut and beans are considered as non-albuminous seeds. Why?
120. Seeds of wheat, maize, barley, castor, coconut and sunflower are considered as albuminous seeds. Why?
121. Give one example for a seed in which the endosperm is completely consumed during the development of embryo.
122. Give one example for a seed in which the endosperm may persist in a mature seed.
123. What is perisperm?
124. What is the residual persistent nucellus of a seed called?
125. Give an example for a plant which produces seeds that has perisperm.
126. What is pericarp?
127. Name the protective wall of fruit.
128. What is a false fruit?
129. Why apple is called a false fruit?
130. What is a true fruit?
131. Give one example for a fleshy fruit.
132. Give one example for a dry fruit.
133. What are parthenocarpic fruits?
134. Give an example for a plant which naturally produces parthenocarpic fruit.
135. How can parthenocarpy be induced?
136. What is apomixis?
137. What is polyembryony?

Two Marks Questions:

1. Name the parts of an angiosperm flower in which development of male and female gametophytes takes place.
2. Mention two differences between microsporogenesis and megasporogenesis.
3. Name the male gametophyte of flowering plants. What is the ploidy of functional microspore in flowering plants?
4. Mention four wall layers of microsporangium.
5. What are the functions of tapetum of microsporangium and filiform apparatus of synergids?
6. What is filiform apparatus? What is its significance?
7. Differentiate syncarpous and apocarpous pistils.
8. What are the functions of integument and nucellus of megasporangium.
9. Define pollination. Mention three types of pollination.
10. Differentiate autogamy and geitonogamy.

11. Differentiate autogamy and xenogamy.
12. Differentiate xenogamy and geitonogamy.
13. What is xenogamy? Mention its importance.
14. Differentiate between chasmogamous and cleistogamous flowers.
15. Mention two advantages of cleistogamy.
16. Mention one advantage and one disadvantage each of cleistogamy.
17. Mention any two characteristic features of flowers that exhibit anemophily.
18. Mention any two characteristic features of flowers that are pollinated by animals.
19. Yucca plant and a species of moth cannot complete their life cycle without each other. Why?
20. Mention any two outbreeding devices in angiosperms to prevent self pollination.
21. What is self-incompatibility? Why self-pollination does not lead to seed formation in self incompatible species?
22. What is artificial hybridisation ? Name the techniques that are employed to achieve this?
23. What is meant by emasculation ? When does a plant breeder employ this technique?
24. What is bagging technique? How is it useful in artificial hybridisation?
25. With regard to artificial hybridization, what do you understand by emasculation and bagging techniques?
26. What do you understand by double fertilization and triple fusion?
27. Mention the four stages of embryogeny in a dicotyledonous embryo.
28. Distinguish between albuminous and non – albuminous seeds.
29. Differentiate hypocotyl and epicotyl.
30. Differentiate coleoptile and coleorrhiza.
31. Differentiate integument and testa.
32. Differentiate perisperm and pericarp.
33. What is perisperm? Give an example for a plant that produces seeds with perisperm.
34. Mention two favourable conditions which promote the germination of seeds.
35. Mention two advantages that the seeds offer to angiosperms.
36. What are true fruits? Give an example for a plant that produces false fruit.
37. What are false fruits? Give an example.
38. Why apple is called a false fruit? Which part of the flower also forms a part of the fruit in apple plant?
39. Distinguish between true fruits and false fruits.
40. What are parthenocarpic fruits? Give an example for a plant that produces false fruit.
41. Name the phenomenon of formation of seed without fertilization in angiosperms. Give an example for a plant which exhibits this phenomenon.
42. What is parthenocarpic fruit? Mention a plant which naturally produces parthenocarpic fruit.
43. What is apomixis and what is its importance?
44. Define polyembryony. Mention an angiosperm which exhibits polyembryony.

Three Marks Questions:

1. Draw a diagrammatic representation of the L.S. of a flower.
2. Mention three differences between microsporogenesis and megasporogenesis.
3. Draw a labeled diagram of transverse section of a young anther.
4. Draw a labeled diagram of an angiosperm ovule.
5. Explain the structure of a mature female gametophyte in flowering plants .
6. Differentiate autogamy, geitonogmy and xenogamy.
7. Differentiate chasmogamous and cleistogamous flowers. How is cliestogamy advantageous ?
8. Mention any three characteristic features of flowers that are pollinated by animals.
9. List three characters of insect pollinated flowers.

10. List three characters of wind pollinated flowers.
11. Explain briefly pollination in Vallisneria.
12. Explain briefly pollination in Zostera.
13. Briefly describe three outbreeding devices in flowering plants.
14. What is meant by emasculation? When does a plant breeder employ this technique and why?
15. Write a short note on pollen-pistil interaction.
16. What is triple fusion? Where does it take place in the megasporangium? Name the nuclei involved in triple fusion.
17. Draw a labeled diagram of a typical dicot embryo.
18. Draw a labeled diagram of the L.S. of an embryo of grass.
19. Explain the structure of a typical dicot embryo.
20. Explain the structure of a typical monocot embryo.
21. Differentiate albuminous and non – albuminous seeds with an example for each.
22. Mention three advantages offered by the seeds to angiosperms.
23. Draw a diagram of the section of fruit in apple.

Five Marks Questions:

1. Explain the T.S. of a microsporangium with a labeled diagram.
2. Describe the structure of a mature male gametophyte.
3. Describe the structure of an anatropous ovule with a labeled diagram.
4. Explain the structure of a typical dicot embryo with a labeled diagram.
5. Explain the structure of a typical monocot embryo with a labeled diagram.
6. Mention five differences between microsporogenesis and megasporogenesis.
7. Explain the development of female gametophyte in angiosperms.
8. Explain how some plants are adapted for achieving pollination through wind.
9. How Vallisneria and Zostera achieve pollination?
10. What is autogamy ? Explain the devices that the plants have developed to prevent this
11. With a neat diagram, explain the structure of a mature female gametophyte of angiosperms.
12. Why plants have developed devices to discourage self pollination? Explain the devices that flowering plants have developed to encourage cross pollination.
13. Give the definitions of the following
(a) Perisperm (b) Pericarp (c) Parthenocarpic fruits (d) Apomixis (e) Polyembryony